

Radar_Sensor_Preprocessing Unit Test Report

Version 0.1.0

Summary

Tests	Failures	Errors	Success rate	Time
165	0	0	100.00%	0.021

Note: *failures* are anticipated and checked for with assertions while *errors* are unanticipated.

Packages

Note: package statistics are not computed recursively, they only sum up all of its testsuites numbers.

Name	Tests	Errors	Failures	Time(s)	Time Stamp
	165	0	0	0.021	2024-03-12T10:13:54

Package

Name	Tests	Errors	Failures	Time(s)	Time Stamp
compute_detection_properties	4	0	0	0.000	2024-03-12T10:13:54
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation	11	0	0	0.001	2024-03-12T10:13:54
f360_check_det_az_conf_and_elevation	1	0	0	0.000	2024-03-12T10:13:54
f360_clear_dets_vcs_long_sorted_info	1	0	0	0.000	2024-03-12T10:13:54
f360_detection_multipath_filter	6	0	0	0.001	2024-03-12T10:13:54
f360_ego_motion_compensate_dets__moving_forward_host_qualtest	1	0	0	0.000	2024-03-12T10:13:54
f360_ego_motion_compensate_dets__stopped_host_qualtest	1	0	0	0.000	2024-03-12T10:13:54
f360_filter_detections_with_bad_azimuth_confidence	1	0	0	0.000	2024-03-12T10:13:54
f360_get_first_relevant_long_sorted_det_idx	3	0	0	0.003	2024-03-12T10:13:54
f360_host_props_update	12	0	0	0.000	2024-03-12T10:13:54
f360_inputs_preprocessing	1	0	0	0.000	2024-03-12T10:13:54
f360_math_func	41	0	0	0.000	2024-03-12T10:13:54
f360_math_func_f360_rotate_2d_vector	4	0	0	0.000	2024-03-12T10:13:54
f360_norm_heading_angle	1	0	0	0.000	2024-03-12T10:13:54
f360_sensor_capability	3	0	0	0.004	2024-03-12T10:13:54
f360_sensor_capability_Compute_Raw_Detection_Uncertainty	2	0	0	0.000	2024-03-12T10:13:54
f360_sensor_capability_Get_Host_Velocity_Uncertainty	2	0	0	0.000	2024-03-12T10:13:54
f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate	3	0	0	0.000	2024-03-12T10:13:54
f360_sensor_capability_Sensor_Capability_Detections	1	0	0	0.003	2024-03-12T10:13:54
f360_sensor_capability_Sensor_Capability_Host	1	0	0	0.000	2024-03-12T10:13:54
f360_set_default_flags_raw_detections	1	0	0	0.001	2024-03-12T10:13:54
f360_sort_detections_vcs_long	4	0	0	0.001	2024-03-12T10:13:54
f360_statistical_tests	9	0	0	0.000	2024-03-12T10:13:54
f360_uncertainty_propagation	1	0	0	0.000	2024-03-12T10:13:54
f360_update_dets_vcs_long_ref_sorted_info	4	0	0	0.001	2024-03-12T10:13:54
f360_update_sensor_first_detection_index	2	0	0	0.000	2024-03-12T10:13:54
f360_update_sensors	1	0	0	0.000	2024-03-12T10:13:54
f360_update_sensors_fov	2	0	0	0.000	2024-03-12T10:13:54
f360_update_sensors_time	1	0	0	0.000	2024-03-12T10:13:54
f360_update_sensors_valid_info	4	0	0	0.000	2024-03-12T10:13:54
rspp_calibrations	1	0	0	0.000	2024-03-12T10:13:54
rspp_detection_double_bounce_detector	10	0	0	0.003	2024-03-12T10:13:54
rspp_discard_detections_outside_sensor_fov	4	0	0	0.001	2024-03-12T10:13:54
rspp_get_version	1	0	0	0.000	2024-03-12T10:13:54
rspp_input_sanity_check	8	0	0	0.002	2024-03-12T10:13:54
rspp_try_to_dealiase_range_rate	12	0	0	0.000	2024-03-12T10:13:54

TestCase compute_detection_properties

Name	Status	Type	Time(s)
Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_negative_polarity	Success		0.000
Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_positive_polarity	Success		0.000
Test_Updated_detection_Time_with_data	Success		0.000
Compute_Detection_VCS_Coordinates	Success		0.000

TestCase f360_check_det_az_conf_and_elevation

Name	Status	Type	Time(s)
FunctionToTest_Descriptive_Tag	Success		0.000

TestCase f360_clear_dets_vcs_long_sorted_info

Name	Status	Type	Time(s)
Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero	Success		0.000

TestCase f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation

Name	Status Type	Time(s)
calculate_range_rate_comp_at_lower_bound	Success	0.000
calculate_range_rate_comp_at_upper_bound	Success	0.000
calculate_range_rate_comp_if_infinite	Success	0.000
calculate_range_rate_comp_above_interval_width	Success	0.000
calculate_range_rate_comp_below_interval_width	Success	0.001
calculate_range_rate_comp_within_interval_width	Success	0.000
Compute_Detections_Properties_Based_On_Ego_Motion__low_threshold_reversing_host	Success	0.000
Compute_Detections_Properties_Based_On_Ego_Motion__low_speed_high_curvature	Success	0.000
Compute_Detections_Properties_Based_On_Ego_Motion__high_speed_low_curvature	Success	0.000
Compute_Detections_Properties_Based_On_Ego_Motion__low_azimuth_conf	Success	0.000
Compute_Detections_Properties_Based_On_Ego_Motion__low_threshold	Success	0.000

TestCase f360_detection_multipath_filter

Name	Status Type	Time(s)
Detection_Multipath_Filter__Not_SRR5	Success	0.000
Detection_Multipath_Filter__Host_Slow	Success	0.000
Detection_Multipath_Filter__Num_Valid_Dets_Below_Threshold	Success	0.001
Detection_Multipath_Filter__Filter_Num_Bad_Dets_Below_Thresh	Success	0.000
Detection_Multipath_Filter__Filter_Num_Bad_Dets_Above_Thresh	Success	0.000
Detection_Multipath_Filter__Filter_First_Det_Idx_Invalid	Success	0.000

TestCase f360_ego_motion_compensate_dets__moving_forward_host_qualtest

Name	Status	Type	Time(s)
Detection_Motion_Classification__det_rr_comp_calculation	Success		0.000

TestCase f360_ego_motion_compensate_dets__stopped_host_qualtest

Name	Status	Type	Time(s)
Ego_Motion_Compensate_Dets__det_rr_comp	Success		0.000

TestCase f360_filter_detections_with_bad_azimuth_confidence

Name	Status	Type	Time(s)
FunctionToTest_Descriptive_Tag	Success		0.000

TestCase f360_get_first_relevant_long_sorted_det_idx

Name	Status Type	Time(s)
Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_Desired_Value_Full_Loop	Success	0.002
Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_Desired_Value	Success	0.001
Test_Get_First_Relevant_Long_Sorted_Det_Idx_General_Case	Success	0.000

TestCase f360_host_props_update

Name	Status Type	Time(s)
Host_Props_Update_Yawing_Heading_Normalized	Success	0.000
Host_Props_Update_Yawing_Position_Update	Success	0.000
Host_Props_Update_Yawing_Covariance_Update	Success	0.000
Host_Props_Update_Yawing_Heading_Update	Success	0.000
Host_Props_Update_Non_Zero_Sideslip_Position_Update	Success	0.000
Host_Props_Update_Non_Zero_Sideslip_Covariance_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_Y_Position_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_Y_Covariance_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_Y_Heading_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_X_Position_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_X_Covariance_Update	Success	0.000
Host_Props_Update_Driving_Straight_Along_X_Heading_Update	Success	0.000

TestCase f360_inputs_preprocessing

Name	Status	Type	Time(s)
Sensor_Capability_Host__SCM_Yaw_Rate_Std	Success		0.000

TestCase f360_math_func

Name	Status	Type	Time(s)
Test_F360_Translate_2D_Position	Success		0.000
Test_F360_Multiply_2D_Matrix_With_Constant	Success		0.000
Test_Rotate_2D_Covariance_Matrix	Success		0.000
Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3	Success		0.000
Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5	Success		0.000
Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3	Success		0.000
Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle	Success		0.000
Test_F360_Mod_divisor_without_decimals	Success		0.000
Test_F360_Mod_divisor_with_decimals	Success		0.000
Test_F360_matmul_2x2_2x1	Success		0.000
Test_F360_Accumulate_Probabilities	Success		0.000
Test_F360_Power_Scale_Prob	Success		0.000
Test_F360_Safe_Logf	Success		0.000
Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas	Success		0.000
Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas	Success		0.000
Test_F360_Low_Pass_Filter_Angle_First_Order	Success		0.000
Test_F360_Low_Pass_Filter_First_Order	Success		0.000
F360_Linear_Equation_With_Saturation_small_den	Success		0.000
Test_F360_Linear_Equation_With_Saturation	Success		0.000
Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints	Success		0.000
Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints	Success		0.000
Test_F360_2d_Matrix_Determinant	Success		0.000
F360_Linear_Equation_small_den	Success		0.000
F360_Linear_Equation	Success		0.000
F360_Sort_dsc	Success		0.000
F360_Sort_stability	Success		0.000
F360_Sort_no_sort	Success		0.000
F360_Sort_partial	Success		0.000
F360_Sort_sorted_values_not_modified	Success		0.000
F360_Sort_asc	Success		0.000
F360_Sort_success	Success		0.000
F360_Sort_fail	Success		0.000
F360_Saturate	Success		0.000
F360_Matmul_MxN_NxP_Transpose	Success		0.000
F360_Matmul_MxN_NxP	Success		0.000
test_f360_get_hypotenuse_squared_zero_input	Success		0.000
test_f360_get_hypotenuse_squared_positive_input	Success		0.000
test_f360_get_hypotenuse_squared_negative_input	Success		0.000
test_f360_get_hypotenuse_zero_input	Success		0.000
test_f360_get_hypotenuse_positive_input	Success		0.000
test_f360_get_hypotenuse_negative_input	Success		0.000

TestCase f360_math_func_f360_rotate_2d_vector

Name	Status Type	Time(s)
Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90	Success	0.000
Test_F360_Rotate_2D_Vector_Angle_Is_90	Success	0.000
Test_F360_Rotate_2D_Vector_Angle_Is_180	Success	0.000
Test_F360_Rotate_2D_Vector_Angle_Is_Zero	Success	0.000

TestCase f360_norm_heading_angle

Name	Status	Type	Time(s)
FunctionToTest_Descriptive_Tag	Success		0.000

TestCase f360_sensor_capability

Name	Status	Type	Time(s)
Test_Compute_Detection_Cross_Covariances	Success		0.000
Get_Uncertainty_Of_Detection_Position	Success		0.000
Test_Compute_Raw_Host_Speed_Uncertainty	Success		0.004

TestCase f360_sensor_capability_Compute_Raw_Detection_Uncertainty

Name	Status	Type	Time(s)
Test_Compute_Raw_Detection_Uncertainty_2	Success		0.000
Test_Compute_Raw_Detection_Uncertainty	Success		0.000

TestCase f360_sensor_capability_Get_Host_Velocity_Uncertainty

Name	Status	Type	Time(s)
Test_with_backward_speed	Success		0.000
Test_with_Forward_speed	Success		0.000

TestCase f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate

Name	Status	Type	Time(s)
Test_Cos_Sin_Az_Out_Of_Range	Success		0.000
Test_Cos_Sin_Az_On_Boundary_Of_Range	Success		0.000
Test_Cos_Sin_Az_Inside_Range	Success		0.000

TestCase f360_sensor_capability_Sensor_Capability_Detections

Name	Status	Type	Time(s)
Test_Sensor_Capability_Detections	Success		0.003

TestCase f360_sensor_capability_Sensor_Capability_Host

Name	Status	Type	Time(s)
Test_Sensor_Capability_Host	Success		0.000

TestCase f360_set_default_flags_raw_detections

Name	Status	Type	Time(s)
set_default_values_for_raw_detections	Success		0.001

TestCase f360_sort_detections_vcs_long

Name	Status Type	Time(s)
Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point	Success	0.000
Test_Sort_Detections_Vcs_Long_No_Detection	Success	0.001
Test_Sort_Detections_Vcs_Long_One_Detection	Success	0.000
Test_Sort_Detections_Vcs_Long_General	Success	0.000

TestCase f360_statistical_tests

Name	Status Type	Time(s)
Check_Stationary_Hypothesis__sigma_below_threshold_test_variable_above_threshold	Success	0.000
Check_Stationary_Hypothesis__sigma_above_threshold_test_variable_below_threshold	Success	0.000
Check_Stationary_Hypothesis__sigma_and_test_variable_above_thresholds	Success	0.000
Check_Stationary_Hypothesis__handles_std_equal_to_zero	Success	0.000
F360_Try_To_Reject_Test__properly_does_not_reject_hypothesis	Success	0.000
F360_Try_To_Reject_Test__properly_rejects_hypothesis	Success	0.000
F360_Chi_Square_Test_Two_Degree_P_Value__test_variable_greater_than_max_test_variable	Success	0.000
F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_saturated	Success	0.000
F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_calculated	Success	0.000

TestCase f360_uncertainty_propagation

Name	Status	Type	Time(s)
Test_Constant_Uncertainty_Propagation_2d	Success		0.000

TestCase f360_update_dets_vcs_long_ref_sorted_info

Name	Status Type	Time(s)
Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_All_Ref_Points	Success	0.000
Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_All_Ref_Points	Success	0.000
Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero_Prev_Ref_Point_Exist	Success	0.001
Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero	Success	0.000

TestCase f360_update_sensor_first_detection_index

Name	Status	Type	Time(s)
update_sensor_first_detection_index_too_many_detections	Success		0.000
Update_Sensor_First_Detection_Index	Success		0.000

TestCase f360_update_sensors

Name	Status	Type	Time(s)
FunctionToTest_Descriptive_Tag	Success		0.000

TestCase f360_update_sensors_fov

Name	Status	Type	Time(s)
update_sensors_fov_azimuth_beyond_fov	Success		0.000
update_sensors_fov	Success		0.000

TestCase f360_update_sensors_time

Name	Status	Type	Time(s)
update_time_since_measurement_in_sensor_props	Success		0.000

TestCase f360_update_sensors_valid_info

Name	Status	Type	Time(s)
check_invalid_sensor	Success		0.000
check_srr_esr_valid	Success		0.000
check_srr_valid	Success		0.000
check_esr_valid	Success		0.000

TestCase rspp_calibrations

Name	Status	Type	Time(s)
FunctionToTest_Descriptive_Tag	Success		0.000

TestCase rspp_detection_double_bounce_detector

Name	Status Type	Time(s)
Test_one_detection_per_sensor	Success	0.001
Test_with_2_detection_not_in_limit_1	Success	0.000
Test_with_2_detection_not_in_limit_2	Success	0.000
Test_with_2_detection_not_in_limit_3	Success	0.000
Test_with_2_detection_inside_gate	Success	0.000
Test_with_2_detection_outside_gate_1	Success	0.001
Test_with_2_detection_outside_gate_2	Success	0.000
Test_with_2_detection_outside_gate_3	Success	0.000
Test_with_2_detection_outside_gate_4	Success	0.000
Test_with_2_detection_with_double_bounce	Success	0.001

TestCase rspp_discard_detections_outside_sensor_fov

Name	Status	Type	Time(s)
Check_Detections_Are_Flagged_As_Per_Sensor_Lookid	Success		0.001
All_Detections_But_One_Outside_FoV	Success		0.000
Check_Detections_Are_Flagged_Based_On_Their_Sensor_Id	Success		0.000
All_Detections_Inside_FoV	Success		0.000

TestCase rspp_get_version

Name	Status	Type	Time(s)
RSPP_Get_Version_Output_Test	Success		0.000

TestCase rspp_input_sanity_check

Name	Status Type	Time(s)
Input_Sanity_Check_Test_With_Sensor_Valid	Success	0.001
Input_Sanity_Check_Test_With_Too_High_Interval_Width	Success	0.000
Input_Sanity_Check_Test_With_Too_Low_Interval_Width	Success	0.001
Input_Sanity_Check_Test_With_Invalid_Sensor_Id	Success	0.000
Input_Sanity_Check_Test_With_Invalid_Detection	Success	0.000
Input_Sanity_Check_Test_With_Two_Detections	Success	0.000
Input_Sanity_Check_Test_With_No_Detection	Success	0.000
Input_Sanity_Check_Test_With_Valid_Detection	Success	0.000

TestCase rspp_try_to_dealiase_range_rate

Name	Status Type	Time(s)
Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_First_Upper	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_Base	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Pred_Exactly_On_Limit	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Between_First_And_Second_Upper_Both_In_Second	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fourth_Lower	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fifth_Lower	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_First_Upper	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_Second_Upper	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Third_Upper_Interval	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Second_Lower_Interval	Success	0.000
Try_To_Dealiase_Range_Rate_Test_Base_Interval	Success	0.000